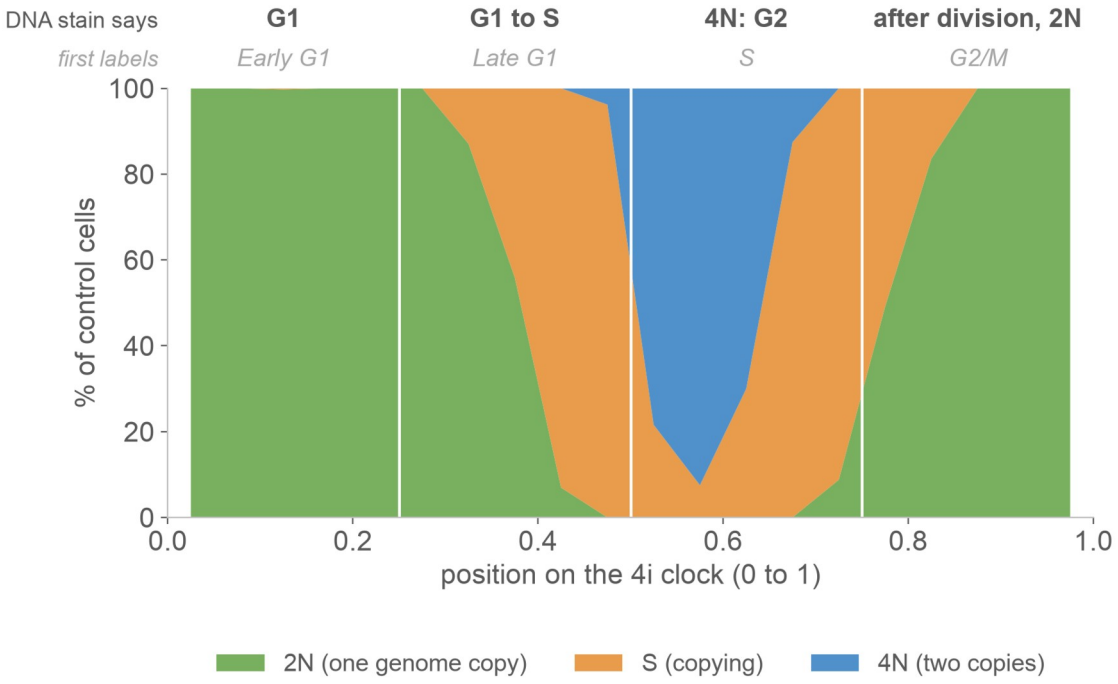
What each part of the 4i clock contains



76%

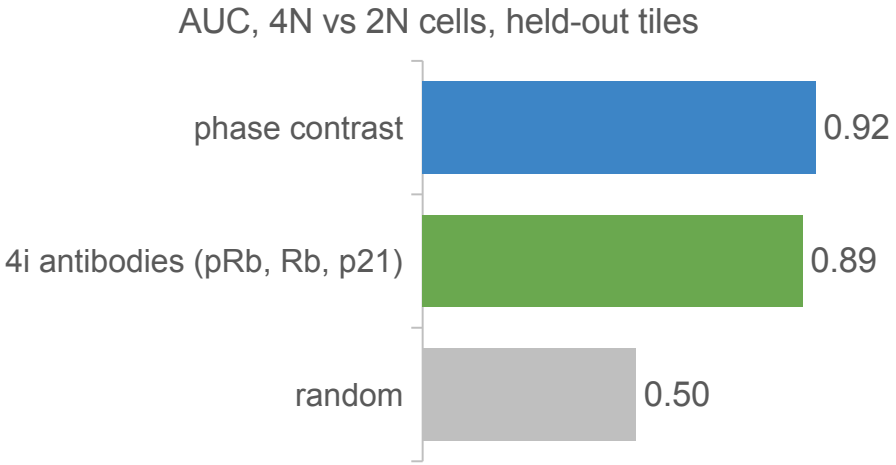
of cells at 0.5 to 0.75 have doubled DNA (4N)

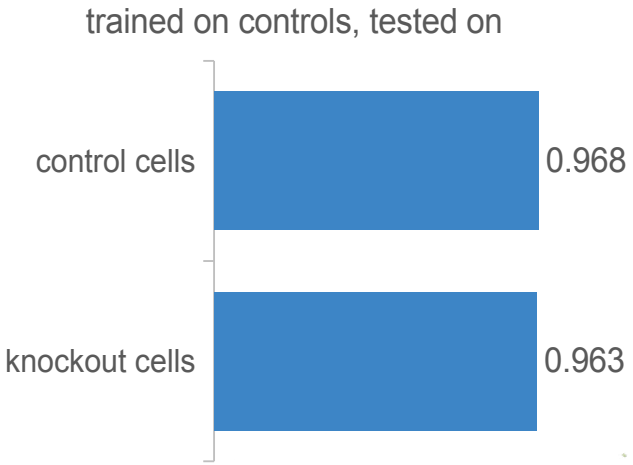
0.75 to 1

is cells just after division, back to 2N

So the first labels were a quarter turn off.

Phase contrast reads DNA state like the antibody panel

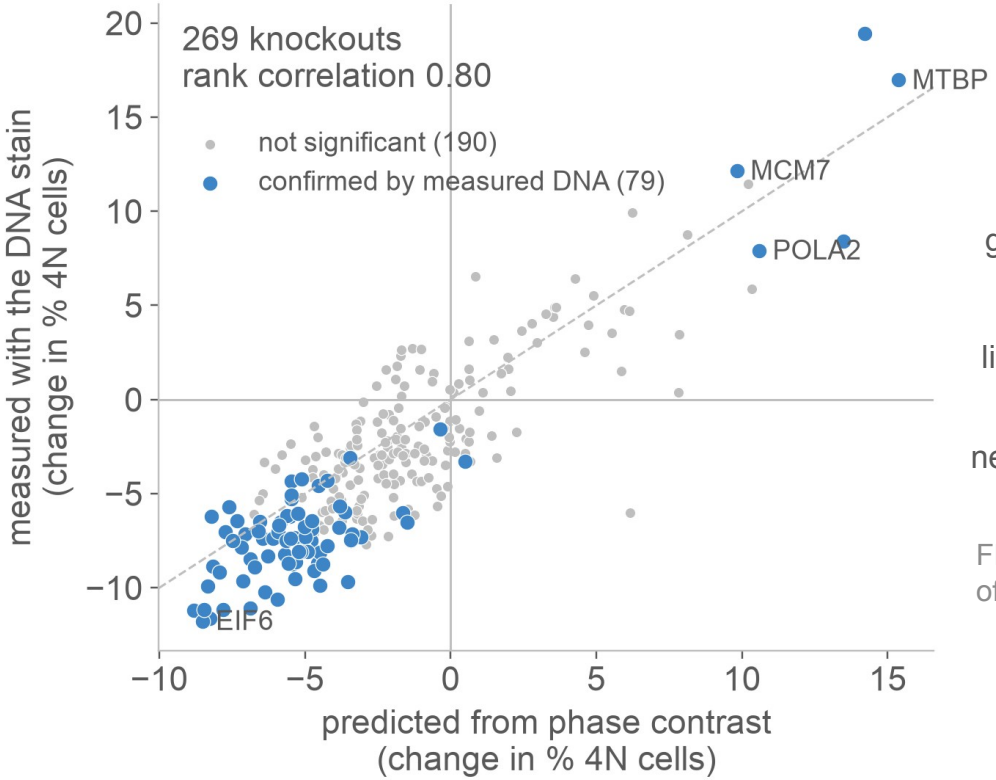


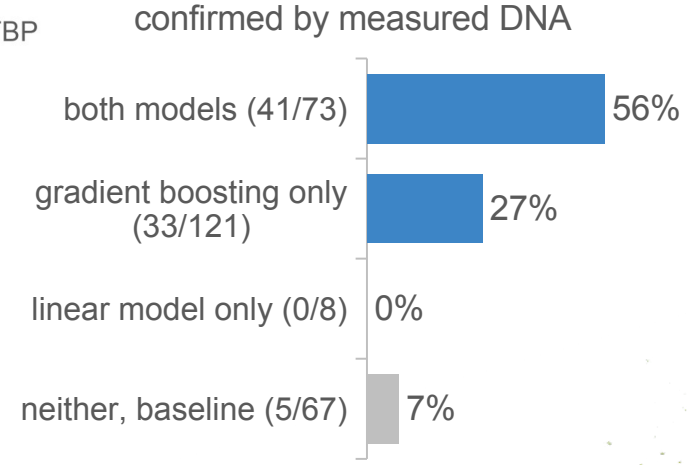


AUC: the chance that a random 4N cell scores higher than a random 2N cell (0.5 is chance, 1 is perfect).

Any brightfield or phase-contrast screen can get a cell-cycle readout without spending a fluorescence channel.

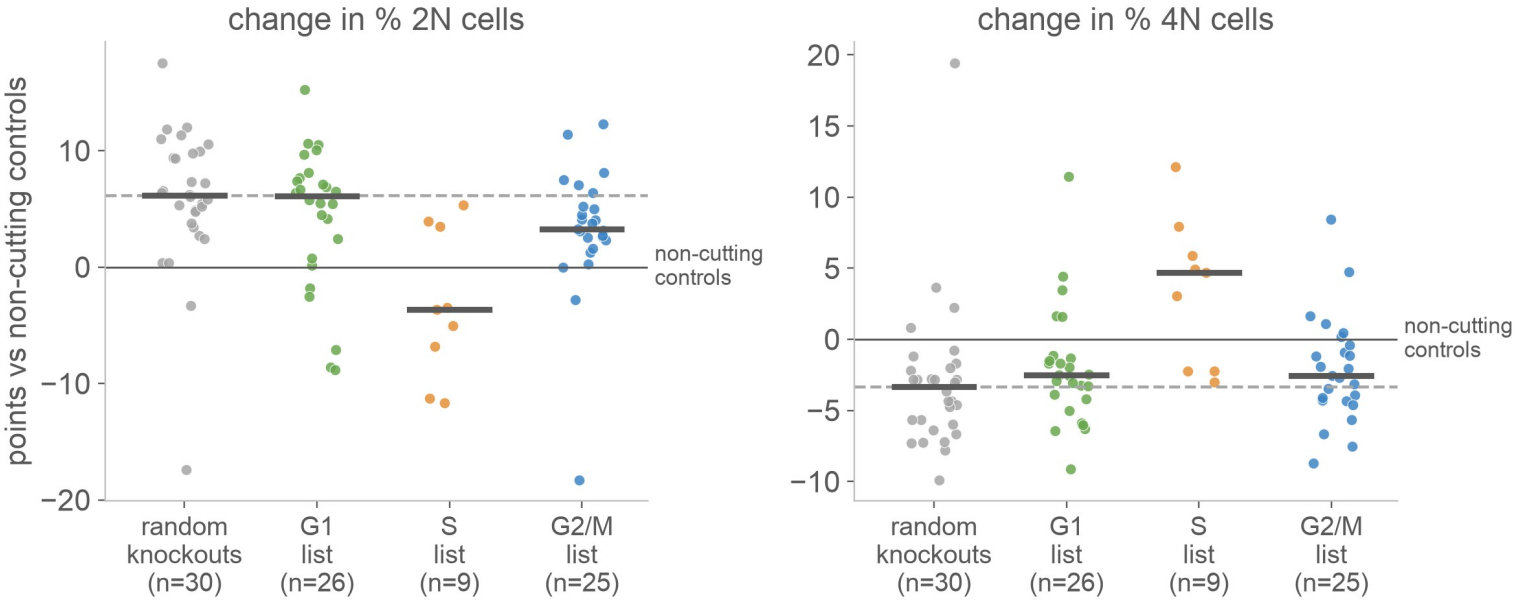
Predicted shifts hold up against measured DNA





FDR 5% against control groups
of the same size.

Judge expected arrests against random knockouts

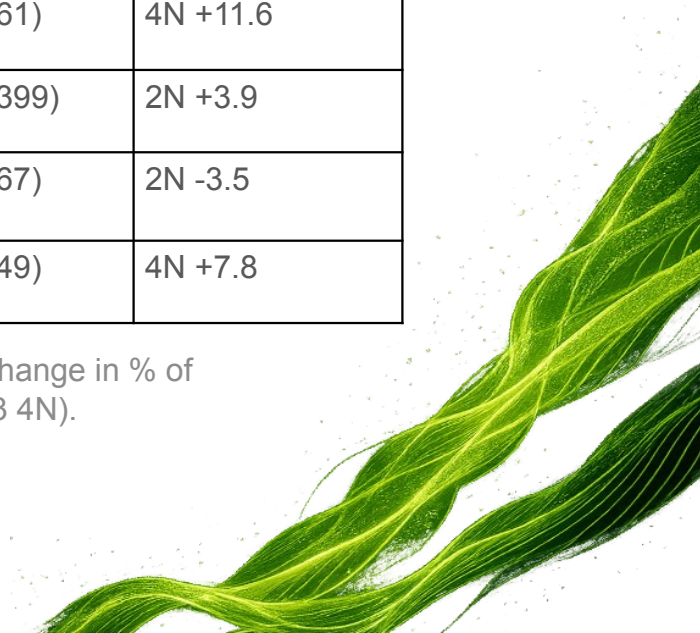


Dashed line: median of 30 random knockouts. S-phase list passes; G1 and G2/M lists look like random knockouts.

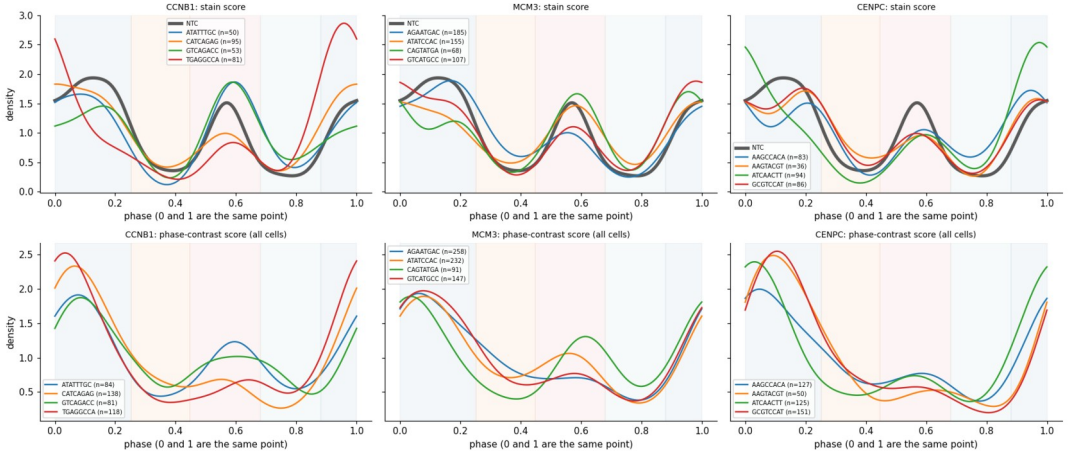
Brian's updated panels against measured DNA

knockout	Brian's expectation	clock shift (panel)	measured DNA vs controls	vs random knockouts
AURKB	cytokinesis failure: G2	G2/M +7.1 (Z 3.2)	4N +3.6 (n=335)	4N +7.0
POLA1	fork stalling: S or G2/M	G2/M +10.4 (Z 5.4)	4N +6.9 (n=457)	4N +10.2
POLA2	fork stalling: S or G2/M	G2/M +14.7 (Z 6.0)	4N +8.2 (n=261)	4N +11.6
CDK7	G1 restriction point: G1	early G1 +8.5 (Z 2.9)	2N +10.0 (n=399)	2N +3.9
BUB1B	checkpoint loss: G1 (slippage)	early G1 +6.3 (Z 2.1)	2N +2.6 (n=267)	2N -3.5
CCNB1	G2 block, back to G1	no clear change	4N +4.4 (n=349)	4N +7.8

Clock shifts read from Brian's panels (against matched control groups). DNA: change in % of cells, all cells of each knockout; random knockouts: median of 30 (+6.1 2N, -3.3 4N).



Measured DNA reproduces Louise's CCNB1 guide split



Louise: cells of each guide along the clock (grey: controls). Top: stain score; bottom: phase contrast.

CCNB1 guides, measured change in % 4N cells: ATATTTGC +14, GTCAGACC +11, CATCAGAG +1, TGAGGCCA -2.

The usual shuffled-cell test calls 12% of control-guide sets different (5% expected); compare guides with control guides instead.

gene	p, shuffled cells	p, vs control guides
CCNB1	0.227	0.352
MCM3	0.033	0.131
CENPC	0.783	0.869
AURKB	0.001	0.004
POLA1	0.477	0.626
POLA2	0.001	0.002
BUB1B	0.141	0.216
CDK7	0.817	0.882

Do a gene's 4 guides differ in measured DNA? (all cells)